

Playwright Automation Framework – Strategic Value & Technical Overview For F5 CMSR Program

Executive Summary

The Playwright Automation Framework developed for the F5 CMS Replacement (CMSR) program is a foundational quality solution supporting the continuous evolution of F5.com. It acts as a design validation engine, regression accelerator, and scalable automation platform that reduces release risk, accelerates delivery, and enforces brand consistency across CMS-managed pages.

Why Playwright Is Essential for the CMS Program

CMSR Requires High Accuracy and Fast Testing Cycles

CMS UI changes frequently due to design updates, content modifications, breakpoint adjustments, component updates, and A/B testing. Manual testing becomes slow and inconsistent. Playwright provides: • High-speed execution • Multi-browser coverage • Auto-waiting to eliminate flaky results • Visual validation ensuring design quality

Playwright Supports All CMS Requirements Out-of-the-Box

• Pixel-accurate FE → Screenshot + design comparison → Ensures Figma consistency • Multi-browser support → Chromium, WebKit, Firefox → Matches real customer usage • Responsive behavior → Device emulation → Validates mobile scenarios • Rapid regression → Parallel execution → Minutes instead of hours • CI integration → ADO-ready → Automated PR checks • Visual hierarchy validation → Structural locator checks → Prevents UI breakage

Framework Features That Directly Benefit CMSR

Figma → JSON → Playwright Validation

Expected UI values are extracted from Figma into JSON, and Playwright compares live UI to design definitions. Benefits: • Removes manual visual checking • Ensures design consistency • Prevents

layout regressions • Reduces QA + design review cycles

Page Object Model (POM) Architecture

POM ensures scalability, maintainability, and clean test structure.

JSON-Driven Expected Result Layer

Each CMS page has JSON defining labels, KPIs, hierarchy, and component visibility. Updating JSON instantly updates validations without modifying test logic.

Azure DevOps (ADO) Integration

CI Quality Gates

Automated PR smoke tests, visual checks, and layout validation before merges.

Nightly Regression

Automated multi-browser matrix testing with screenshot and JSON validation.

Artifacts Uploaded Automatically

Screenshots, trace logs, console logs, network logs, and videos are automatically stored in ADO for debugging.

Reporting, Screenshots & Failure Analysis

Automatic Screenshots

Every failure captures a screenshot, locator highlight, and expected vs actual comparison.

Trace Viewer

Provides DOM snapshots, network logs, console logs, and full timeline replay.

Video Recording

Reveals flicker, animation issues, and browser rendering inconsistencies.

Rich HTML Reports

Includes execution times, embedded screenshots, trace links, and browser matrix results.

JSON Comparison Logs

Highlights differences between Figma-derived expectations and actual UI rendering.

Mobile & Responsive Testing Support

Device Emulation

Supports iPhone, Android, and tablets. Validates layout shifts, collapses, and navigation behavior.

Cloud Device Testing

BrowserStack or Sauce Labs can be integrated without modifying any tests.

Business Impact of Adopting Playwright

Faster Delivery

Regression time reduces from days to minutes, enabling rapid iteration.

Improved Design Quality

Ensures brand consistency and prevents UI regressions.

Lower Long-Term QA & Dev Costs

Automation eliminates repeated manual testing, reducing cost and human error.

Strong Operational Stability

Screenshots, traces, and videos accelerate debugging and reduce production risks.

Supports the Future CMS Vision

Supports API+UI flows, mobile expansion, CI-driven automation, and JSON validation.

Roadmap for the Next 3–6 Months

Phase 1 – Expand Test Coverage

Glossary, Background Jobs, Route Summary KPIs, E2E flows.

Phase 2 – ADO Full Integration

Automated PR checks, daily regression, dashboards.

Phase 3 – Mobile Automation

Breakpoint validation, device testing.

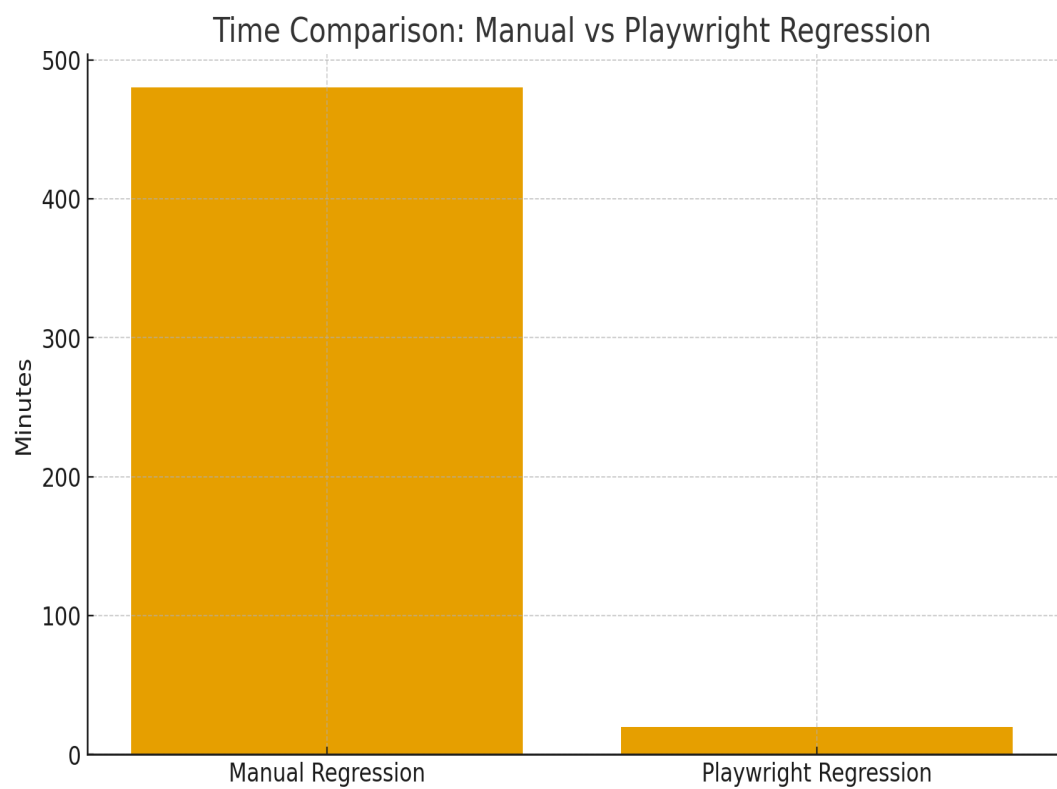
Phase 4 – API + UI Combined Tests

Route computations, KPI validation.

Phase 5 – Accessibility Automation

Axe-core WCAG accessibility validation.

Playwright vs Manual Regression – Time Comparison Chart



This chart shows regression time improvement: • Manual regression: ~480 minutes (8 hours) • Playwright regression: ~20 minutes This represents over a 95% reduction in regression effort and significantly accelerates delivery.